

$$\nabla \cdot \vec{E} = \frac{1}{\epsilon_0}(\rho_f - \nabla \cdot \vec{P}) \Rightarrow \nabla \cdot (\underbrace{\epsilon_0 \vec{E} + \vec{P}}_{\vec{D}}) = \rho_f \Rightarrow$$

$$\boxed{\nabla \cdot \vec{D} = \rho_f}$$

From Ampere's Law with Maxwell's addition:

$$\nabla \times \vec{B} = \mu_0 \left(\vec{J}_f + \nabla \times \vec{M} + \frac{\partial}{\partial t} \vec{P} \right) + \mu_0 \epsilon_0 \frac{\partial}{\partial t} \vec{E} \Rightarrow$$

$$\nabla \times \left(\underbrace{\frac{\vec{B}}{\mu_0} - \vec{M}}_{\vec{H}} \right) = \vec{J}_f + \underbrace{\frac{\partial}{\partial t} (\epsilon_0 \vec{E} + \vec{P})}_{\substack{\vec{D} \rightarrow \text{displacement} \\ \partial_t \vec{D}: \text{displacement current}}} \Rightarrow$$

$$\boxed{\nabla \times \vec{H} = \vec{J}_f + \frac{\partial}{\partial t} \vec{D}}$$

while the rest two equations, $\nabla \cdot \vec{B} = 0$, $\nabla \times \vec{E} = -\partial_t \vec{B}$, remain unchanged.

4. Boundary conditions

We employ the integral form of Maxwell's equations to derive the boundary conditions. In the case of time-dependent electric and magnetic fields, the discontinuities of the field components across boundaries, which we observed in static cases, remain unchanged. This is because the only differences in the dynamic case come from the following two equations:

$$\oint_L \vec{E} \cdot d\vec{\ell} = -\frac{d\Phi_B}{dt}$$

and

$$\oint_L \vec{B} \cdot d\vec{\ell} = \mu_0 \left(I_{\text{enc}} + \epsilon_0 \frac{d\Phi_E}{dt} \right).$$

In static cases, the terms $\frac{d\Phi_B}{dt}$ and $\frac{d\Phi_E}{dt}$ are zero. However, when deriving the boundary conditions using these equations, we reduce the height of the integration loop to zero. As a result, the flux terms Φ_B and Φ_E , as well as $\frac{d\Phi_B}{dt}$ and $\frac{d\Phi_E}{dt}$, also vanish, leading to the same conclusions as we derived in the static case.

5. Poynting's Theorem

Up to now, we have learned how charges and currents generate fields, and how fields exert forces back onto charges. For a deeper understanding of physics, it is important to analyze what happens to electromagnetic energy. This section provides a framework for understanding how electromagnetic energy evolves within a volume V bounded by a closed surface S .

Poynting's Theorem is a fundamental principle in electromagnetism that describes the conservation of energy in the presence of electromagnetic fields. It relates the rate at which electromagnetic energy is transferred or converted within a system to the fields and currents in that system.

Let's begin by analyzing a single point charge in an electromagnetic field. Over a short time interval dt , the work done by the field is:

$$\begin{aligned}\vec{F} \cdot d\vec{\ell} &= q(\vec{E} + \vec{v} \times \vec{B}) \cdot \vec{v} dt \\ &= q\vec{E} \cdot \vec{v} dt.\end{aligned}$$

For a continuous charge distribution, we replace q with ρdV . Here, ρ represents the amount of charge per unit volume, and $\vec{v}$ is the velocity (i.e., displacement per unit time), so the product $\rho\vec{v}$ gives the amount of charge flowing per unit time per unit area—this is precisely the volume current density $\vec{J}$.

Thus, the work done by the electromagnetic field on a volume element dV containing charge density ρ is:

$$\rho\vec{v} \cdot \vec{E} dt dV = \vec{J} \cdot \vec{E} dt dV.$$

Dividing by dt , we obtain the rate at which the electromagnetic field does work on the volume element dV :

$$\vec{J} \cdot \vec{E} dV.$$

To obtain the total electromagnetic work done per unit time, we perform the following volume integral:

$$\frac{dW}{dt} = \int_V \vec{J} \cdot \vec{E} dV.$$

Applying Ampere's Law with Maxwell's addition,

$$\begin{aligned}\int_V \vec{J} \cdot \vec{E} dV &= \int_V \vec{E} \cdot \left(\frac{1}{\mu_0} \nabla \times \vec{B} - \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right) dV \\ &= \underbrace{\frac{1}{\mu_0} \int_V \vec{E} \cdot (\nabla \times \vec{B}) dV}_{\text{Integration by parts}} - \epsilon_0 \int_V \vec{E} \cdot \frac{\partial \vec{E}}{\partial t} dV \\ &= \frac{1}{\mu_0} \int_V \vec{B} \cdot \underbrace{(\nabla \times \vec{E})}_{-\partial_t \vec{B}} dV - \frac{1}{\mu_0} \int_V \nabla \cdot (\vec{E} \times \vec{B}) dV - \epsilon_0 \int_V \vec{E} \cdot \frac{\partial \vec{E}}{\partial t} dV \\ &= -\frac{1}{\mu_0} \int_V \vec{B} \cdot \frac{\partial \vec{B}}{\partial t} dV - \epsilon_0 \int_V \vec{E} \cdot \frac{\partial \vec{E}}{\partial t} dV - \frac{1}{\mu_0} \oint_S (\vec{E} \times \vec{B}) \cdot d\vec{a} \\ &= -\frac{d}{dt} \underbrace{\left[\frac{1}{2\mu_0} \int_V B^2 dV + \frac{\epsilon_0}{2} \int_V E^2 dV \right]}_{\text{Total energy stored in the electromagnetic field}} - \frac{1}{\mu_0} \oint_S (\vec{E} \times \vec{B}) \cdot d\vec{a}\end{aligned}$$

Denote the total energy stored in the electromagnetic field as:

$$U = \frac{1}{2\mu_0} \int_V B^2 dV + \frac{\epsilon_0}{2} \int_V E^2 dV,$$

and denote the vector in the surface integral as:

$$\vec{S} = \frac{1}{\mu_0} \vec{E} \times \vec{B}$$

which is known as the Poynting Vector.

To sum up, we obtain the following equation:

$$\boxed{-\oint_S \vec{S} \cdot d\vec{a} = \frac{dW}{dt} + \frac{dU}{dt}} \xrightarrow{\text{in explicit terms}} \int_V \vec{J} \cdot \vec{E} dV + \frac{d}{dt} \left(\frac{1}{2\mu_0} \int_V B^2 dV + \frac{\epsilon_0}{2} \int_V E^2 dV \right)$$

The equation above is known as Poynting's Theorem in its integral form. It expresses the principle of energy conservation for electromagnetic fields: the electromagnetic energy flowing into a volume (left-hand side) is used in two ways (right-hand side): 1. To do work on matter, converting electromagnetic energy into other forms—such as heat in a resistor, mechanical energy in a motor, or light in a bulb; 2. To be stored in the electric and magnetic fields within the volume. All quantities in the equation represent energy transfer per unit time, or power. The theorem is analogous to the continuity equation, as both describe local conservation of a physical quantity.

According to the above explanation, the Poynting Vector describes the flow of electromagnetic energy in space: its direction indicates the direction in which electromagnetic energy is flowing, whereas the magnitude gives the power per unit area carried by the field — that is, how much energy passes through a unit area per unit time (W/m²).

Let's return to the volume integral form to express all terms explicitly:

$$-\int_V \nabla \cdot \left(\frac{\vec{E} \times \vec{B}}{\mu_0} \right) dV = \int_V \vec{J} \cdot \vec{E} dV + \left[\int_V \frac{\partial}{\partial t} \left(\frac{1}{2\mu_0} B^2 + \frac{\epsilon_0}{2} E^2 \right) dV \right].$$

Since this equation holds for any volume V , the integrands must be equal. This gives the differential form of Poynting's Theorem:

$$\boxed{-\nabla \cdot \vec{S} = \vec{J} \cdot \vec{E} + \frac{\partial u}{\partial t}}.$$

In the above equation, $u = \frac{1}{2\mu_0} B^2 + \frac{\epsilon_0}{2} E^2$ is the electromagnetic energy density. It closely resembles the continuity equation for charge conservation:

$$-\nabla \cdot \vec{J} = \frac{\partial \rho}{\partial t}.$$

In both cases, the negative divergence of a flux equals the rate of change of a density — a hallmark of local conservation laws.